

Department of Computer Science And Engineering (CSE) BRAC University

CSE 251 – Electronic Devices and Circuits

Lecture: 06

Diode

Introduction to semiconductor

PREPARED BY

Tarvir Anjum Aditto (TAV)

Lecturer

Department of CSE, BRAC University

Email: tarvir.anjum@bracu.ac.bd

»»» Intrinsic Semiconductor

- **semiconductor** – a material whose **conductivity lies between** that of conductors (copper) and insulators (glass).
 - **single-element** – such as germanium and silicon.
 - **compound** – such as gallium-arsenide.

»» Intrinsic Semiconductor

- **valence electron** – is an electron that participates in the formation of **chemical bonds**.
 - Atoms with **one or two valence electrons more than a closed shell** are highly reactive because the extra electrons are easily removed to form positive ions.
- **covalent bond** – is a form of chemical bond in which two atoms **share a pair of atoms**.
 - It is a **stable balance of attractive and repulsive forces** between atoms when they share electrons.

»»» Intrinsic Semiconductor

■ silicon atom

- four **valence** electrons
- requires **four more** to complete outermost shell
- each pair of shared forms a **covalent** bond
- the atoms form a **lattice** structure

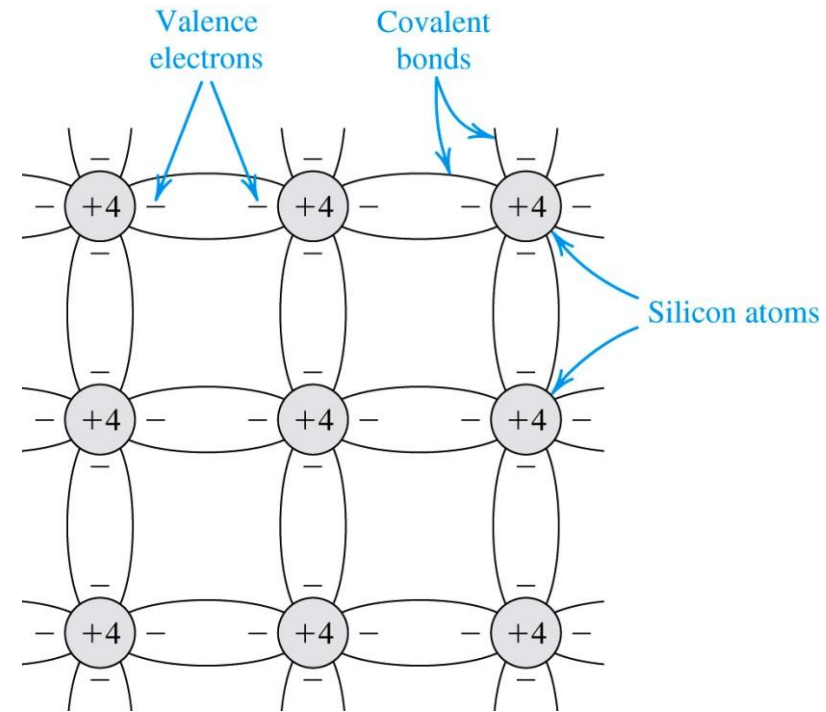


Figure 3.1 Two-dimensional representation of the silicon crystal. The circles represent the inner core of silicon atoms, with +4 indicating its positive charge of $+4q$, which is neutralized by the charge of the four valence electrons. Observe how the covalent bonds are formed by sharing of the valence electrons. At 0K, all bonds are intact and no free electrons are available for current conduction.

»»» Intrinsic Semiconductor

- silicon at **low** temps
 - all **covalent bonds** – are intact
 - no **electrons** – are available for conduction
 - **conductivity** – is zero
- silicon at **room** temp
 - some **covalent bonds** – break, freeing an electron and creating hole, due to thermal energy
 - some **electrons** – will wander from their parent atoms, becoming available for conduction
 - **conductivity** – is greater than zero

»»» Intrinsic Semiconductor

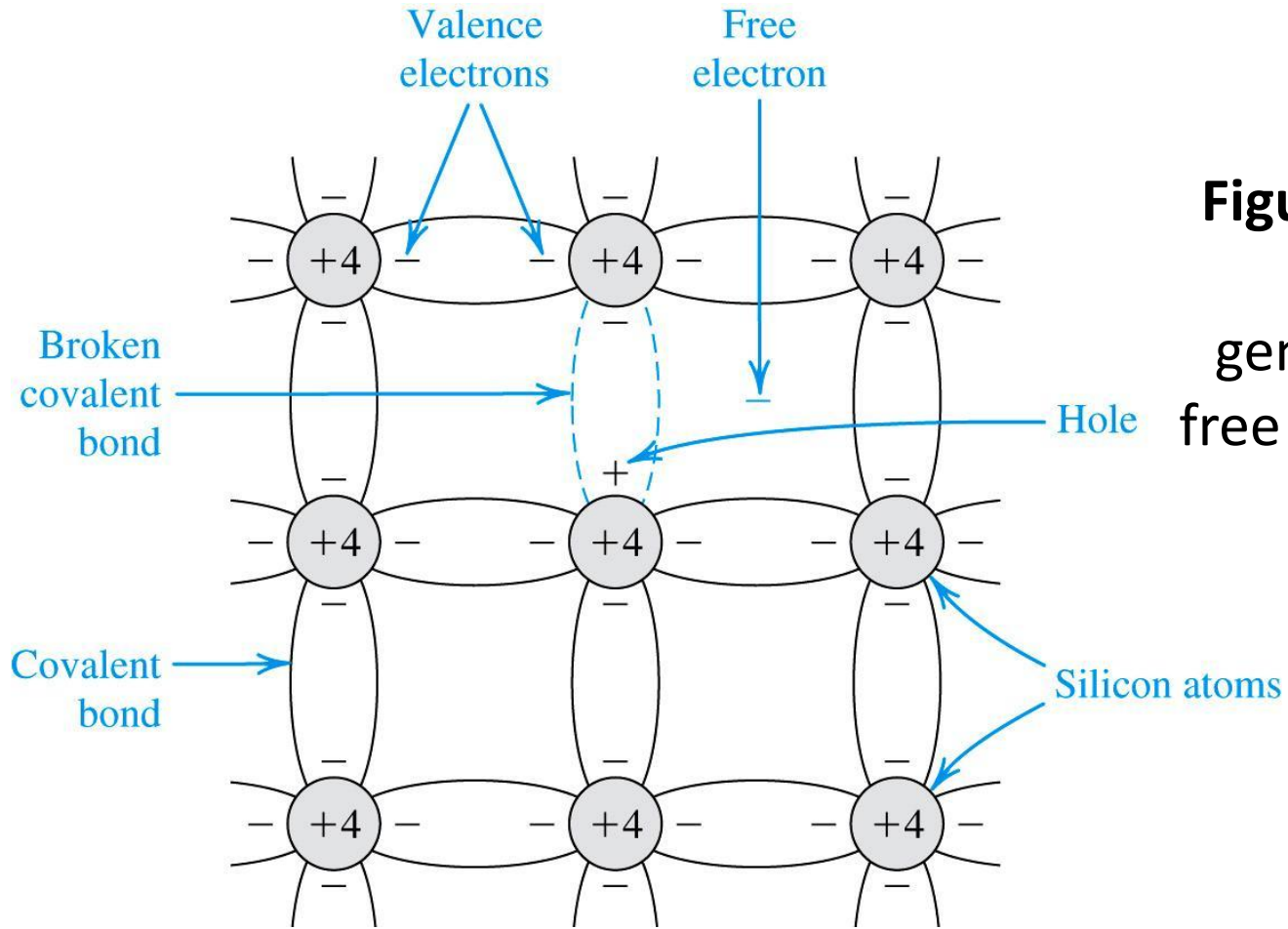


Figure 3.2: At room temperature, some of the covalent bonds are broken by thermal generation. Each broken bond gives rise to a free electron and a hole, both of which become available for current conduction.

»» Intrinsic Semiconductor

- **intrinsic semiconductor** – is one which is **not doped**
 - One example is pure silicon.
- **generation** – is the process of free electrons and holes being **created**.
 - **generation rate** – is speed with which this occurs.
- **recombination** – is the process of free electrons and holes **disappearing**.
 - **recombination rate** – is speed with which this occurs.

Generation may be effected by **thermal energy**. As such, both generation and recombination rates will be (at least in part) a function of temperature.

»» Intrinsic Semiconductor

- **thermal generation** – effects a **equal concentration** of free electrons and holes.
 - Therefore, electrons **move randomly** throughout the material.
- In thermal equilibrium, **generation and recombination rates are equal**.

»» Intrinsic Semiconductor

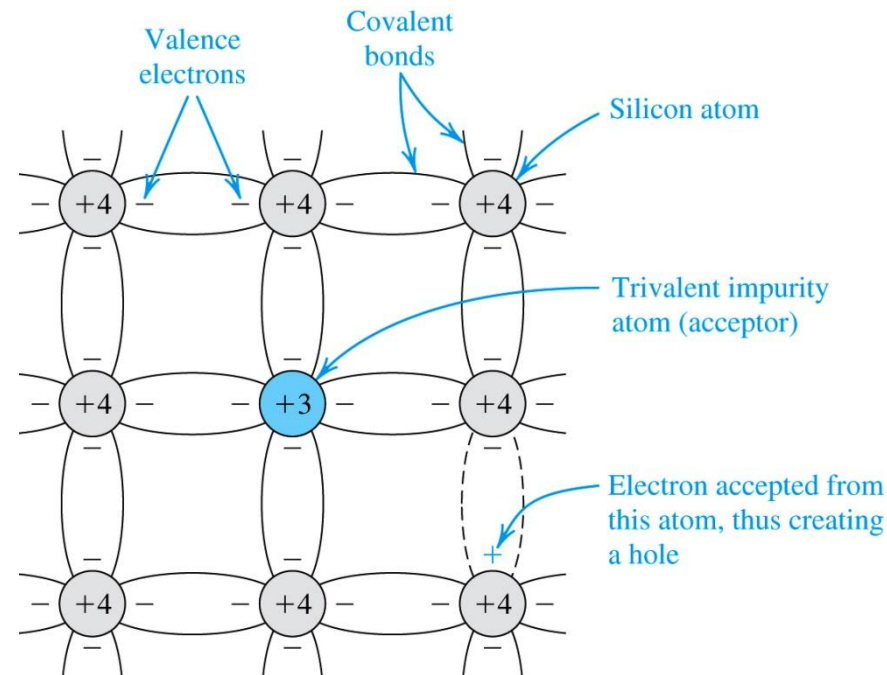
- **Q:** Why can **thermal generation not be used** to effect meaningful current conduction?
 - **A:** Silicon crystal structure described previously is **not sufficiently conductive** at room temperature.
 - Additionally, a **dependence on temperature** is not desirable.
- **Q:** How can this “**problem**” be fixed?
 - **A:** doping

doping – is the **intentional introduction of impurities into an extremely pure (intrinsic) semiconductor** for the purpose changing carrier concentrations.

»»» Doped Semiconductor

■ *p*-type semiconductor

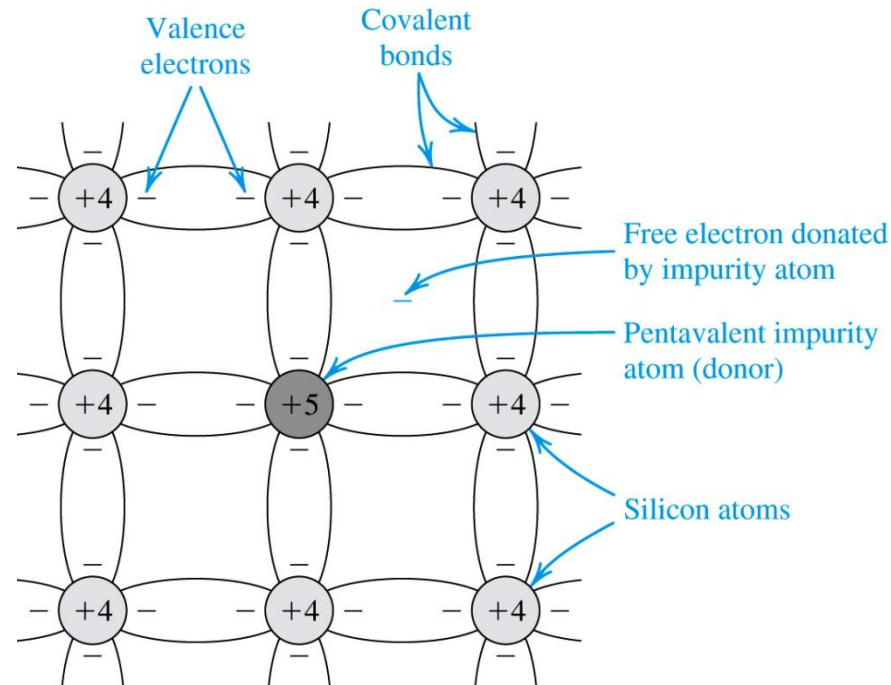
- Silicon is doped with element having a **valence of 3**.
- To increase the concentration of **holes** (*p*).
- One example is **boron**, which is an acceptor.



»»» Doped Semiconductor

■ *n*-type semiconductor

- Silicon is doped with element having a **valence of 5**.
- To increase the concentration of free **electrons** (n).
- One example is **phosphorus**, which is a donor.



»»» Doped Semiconductor

▪ *p*-type semiconductor

- n_p will have the same dependence on temperature as n_i^2
- the concentration of holes (p_n) will be much larger than holes
- holes are the majority charge carriers
- free electrons are the minority charge carrier

▪ *n*-type semiconductor

- p_n will have the same dependence on temperature as n_i^2
- the concentration of free electrons (n_n) will be much larger than holes
- electrons are the majority charge carriers
- holes are the minority charge carrier

»»» Drift Current

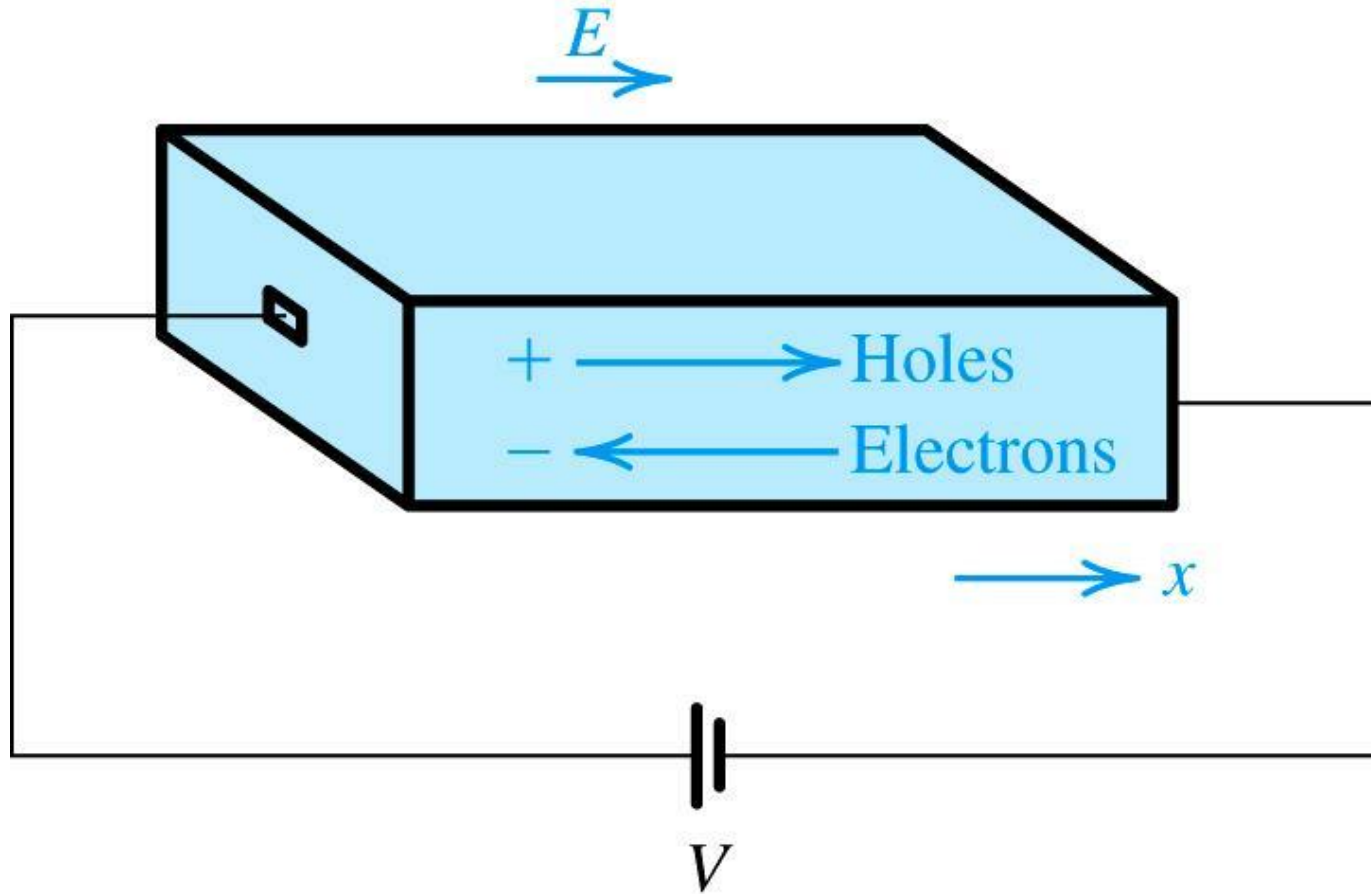


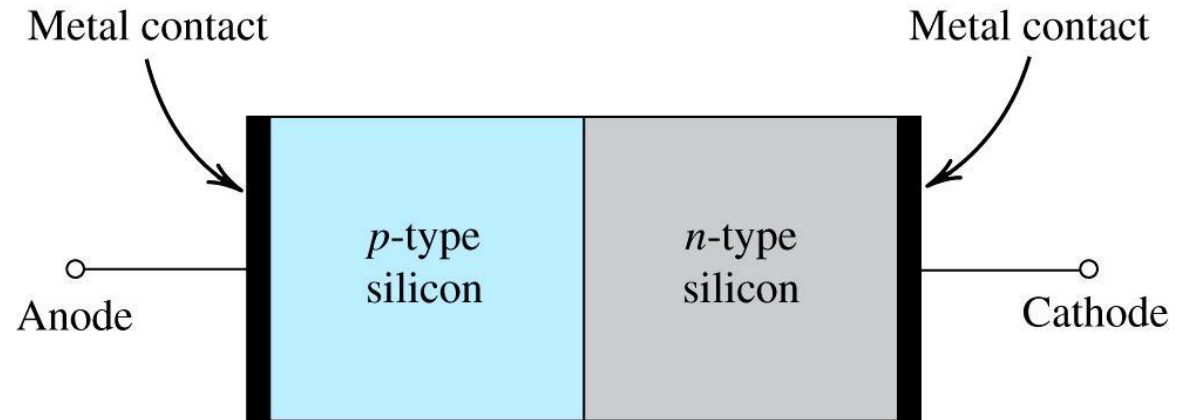
Figure 3.5: An electric field E established in a bar of silicon causes the holes to drift in the direction of E and the free electrons to drift in the opposite direction. Both the hole and electron drift currents are in the direction of E .

»»» Diffusion Current

- **carrier diffusion** – is the flow of charge carriers from area of **high concentration to low concentration**.
 - It requires **non-uniform distribution** of carriers.
- **diffusion current** – is the current flow that results from diffusion.

»»» pn Junction

- **pn junction** structure
 - *p*-type semiconductor
 - *n*-type semiconductor
 - metal contact for connection



»»» pn Junction

- **Q:** What is state of *pn* junction with open-circuit terminals?
- **A:** Read the below...
 - *p*-type material contains majority of holes
 - these holes are neutralized by equal amount of bound negative charge
 - *n*-type material contains majority of free electrons
 - these electrons are neutralized by equal amount of bound positive charge

»» pn Junction

- **bound charge**
 - charge of **opposite** polarity to free electrons / holes of a given material
 - **neutralizes the electrical charge** of these majority carriers
 - does not affect **concentration** gradients

»» pn Junction

- **Q:** What happens when a *pn-junction* is newly formed – aka. when the *p*-type and *n*-type semiconductors first touch one another?
 - **A:** See following slides...

»»» pn Junction

Step #1: The p -type and n -type semiconductors are joined at the junction.

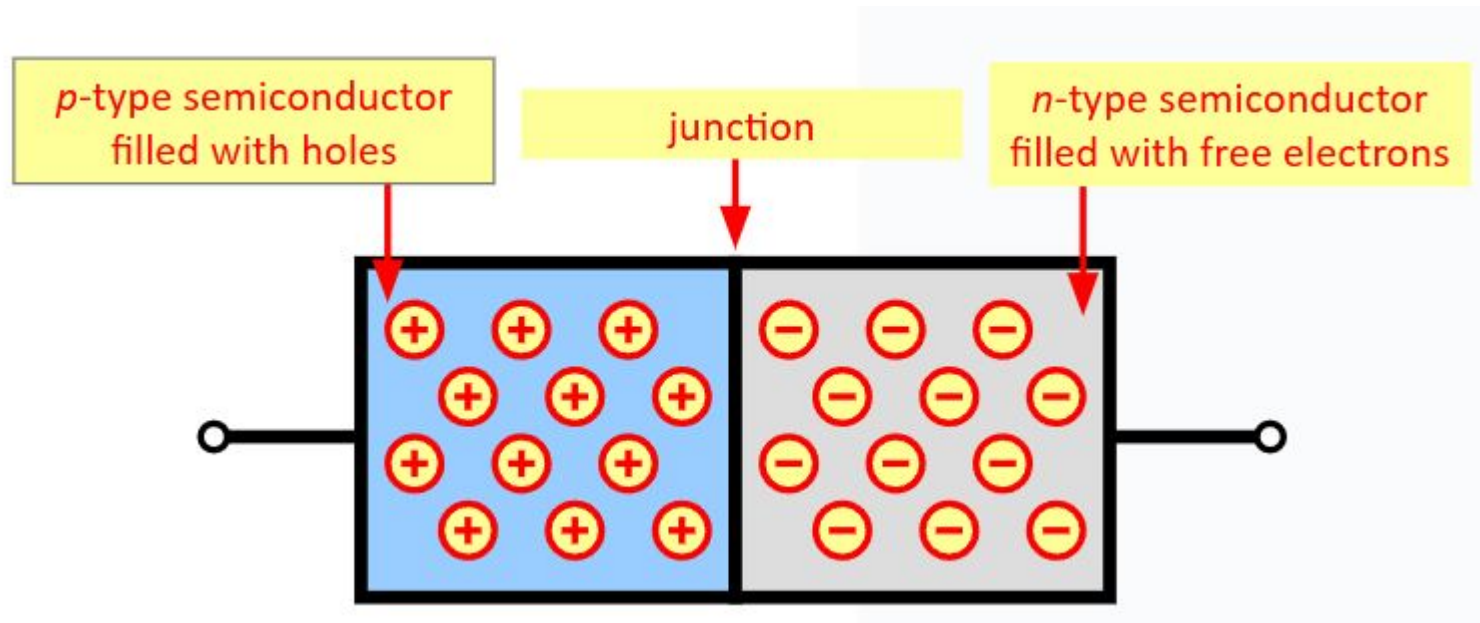


Figure: The pn junction with no applied voltage (open-circuited terminals).

»»» pn Junction

Step #1A: Bound charges are attracted (from environment) by free electrons and holes in the p -type and n -type semiconductors, respectively. They remain weakly “bound” to these majority carriers; however, they **do not recombine**.

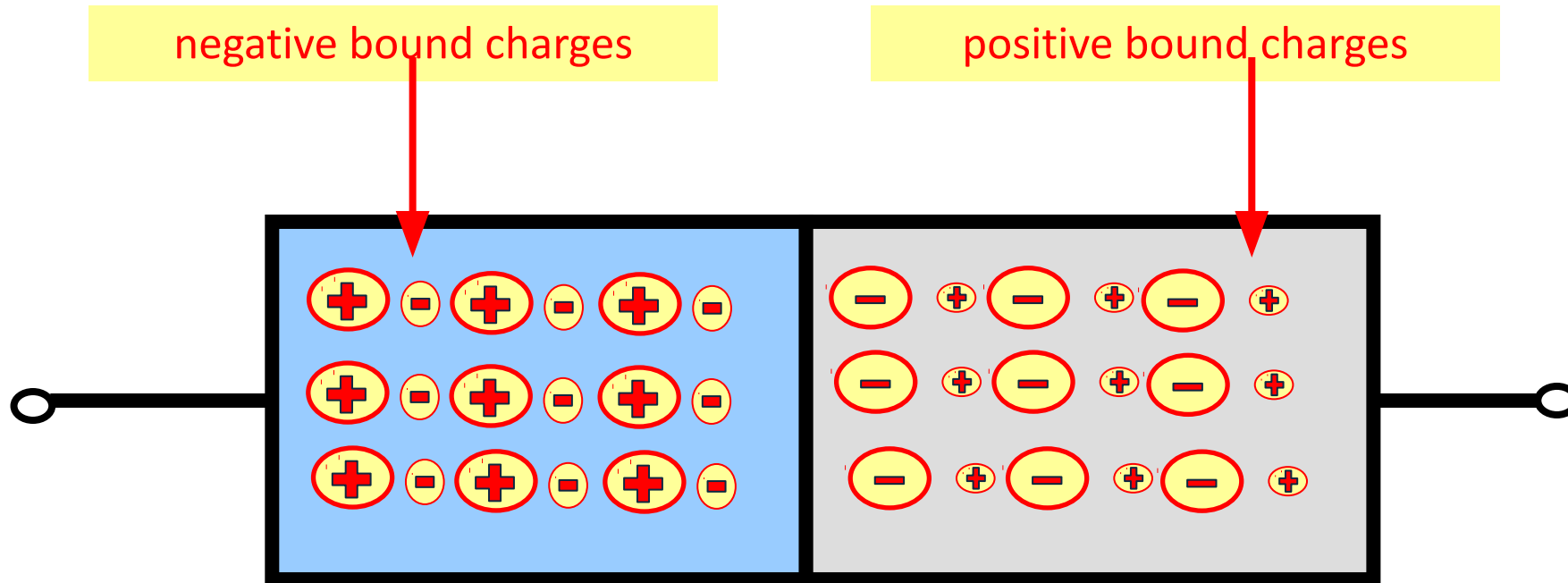


Figure: The pn junction with no applied voltage (open-circuited terminals).

»»» pn Junction

Step #2: Diffusion begins. Those free electrons and holes which are closest to the junction will recombine and, essentially, eliminate one another.

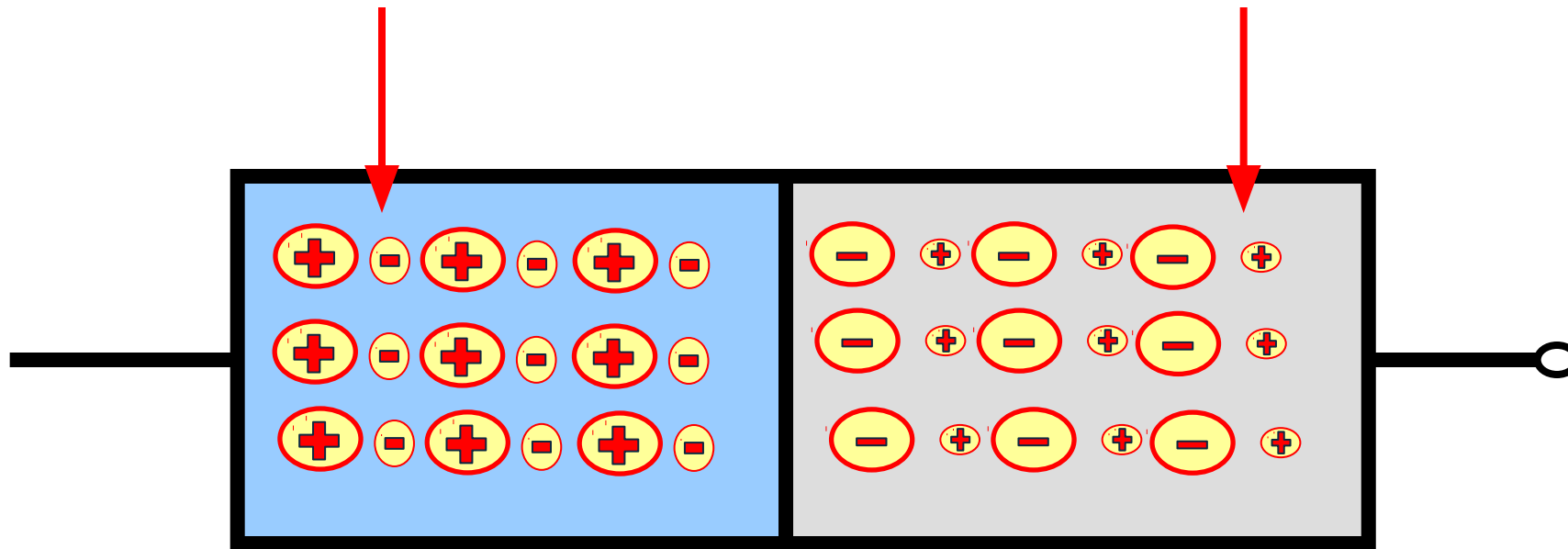


Figure: The *pn* junction with no applied voltage (open-circuited terminals).

»»» pn Junction

Step #3: The **depletion region** begins to form – as diffusion occurs and free electrons recombine with holes.

The depletion region is filled with “uncovered” bound charges – who have lost the majority carriers to which they were linked.

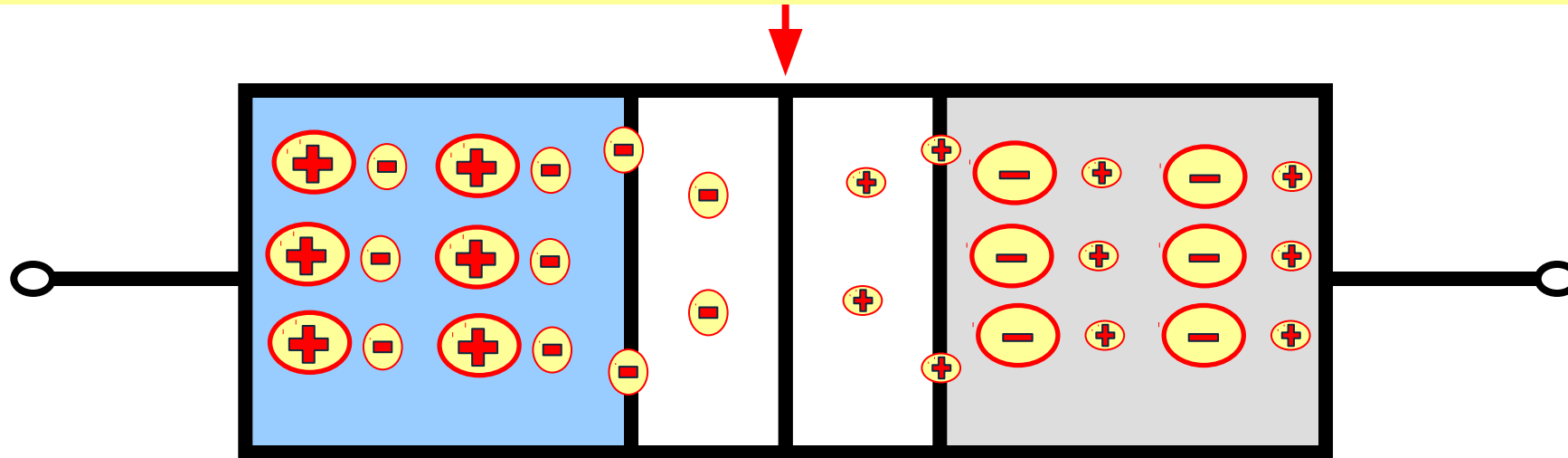
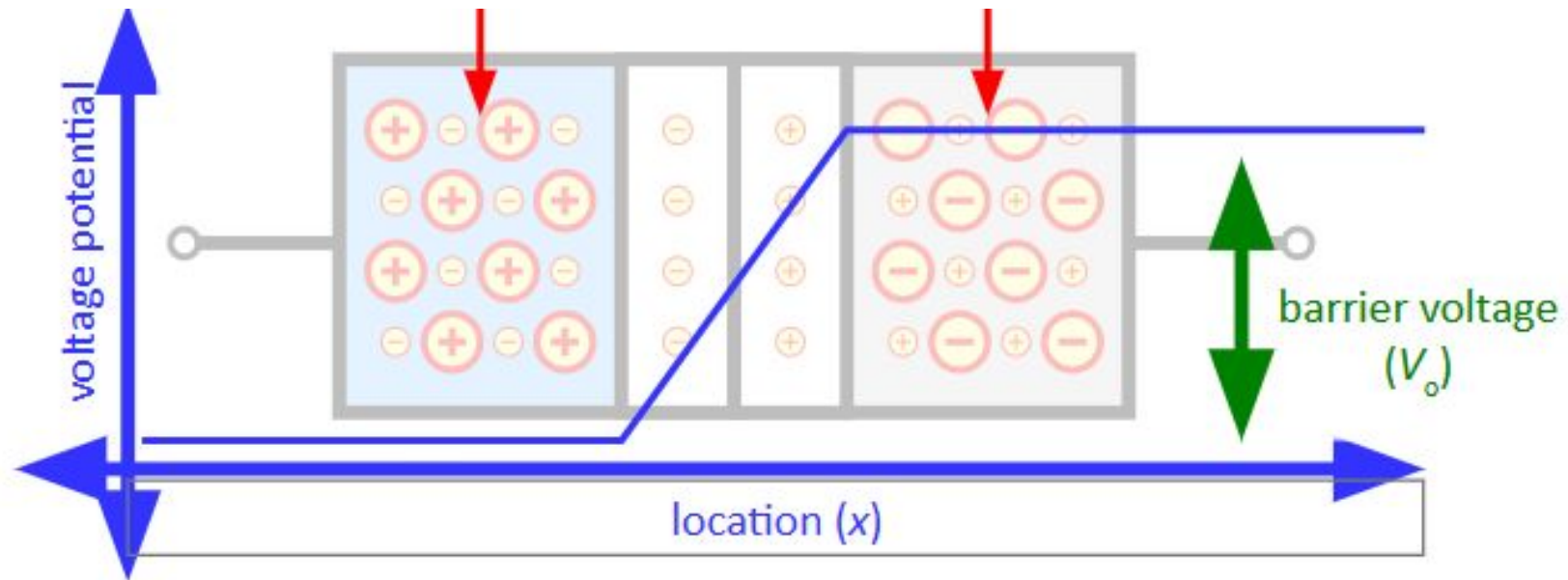


Figure: The *pn* junction with no applied voltage (open-circuited terminals).

»»» pn Junction

Step #4: The “uncovered” bound charges effect a voltage differential across the depletion region. The magnitude of this barrier voltage (V_0) differential grows, as diffusion continues.

No voltage differential exists across regions of the *pn*-junction outside of the depletion region because of the neutralizing effect of positive and negative bound charges.



»»» pn Junction

Step #5: The barrier voltage (V_0) is an electric field whose polarity opposes the direction of diffusion current (I_D). As the magnitude of V_0 increases, the magnitude of I_D decreases.

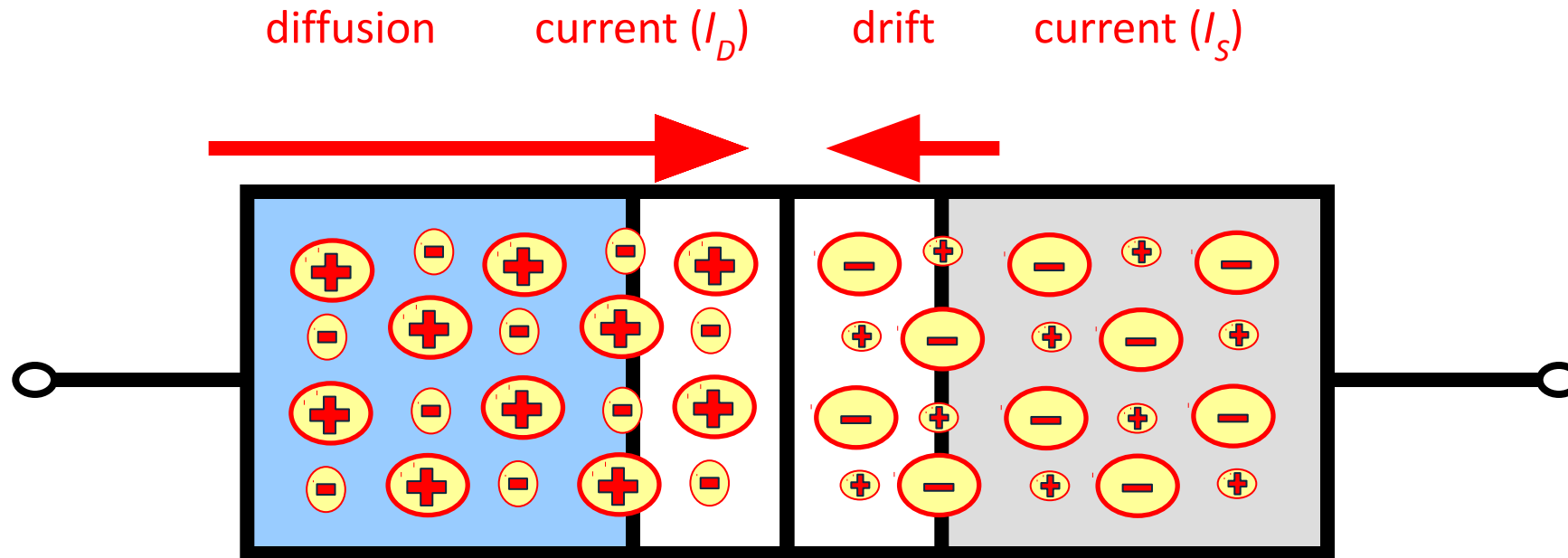
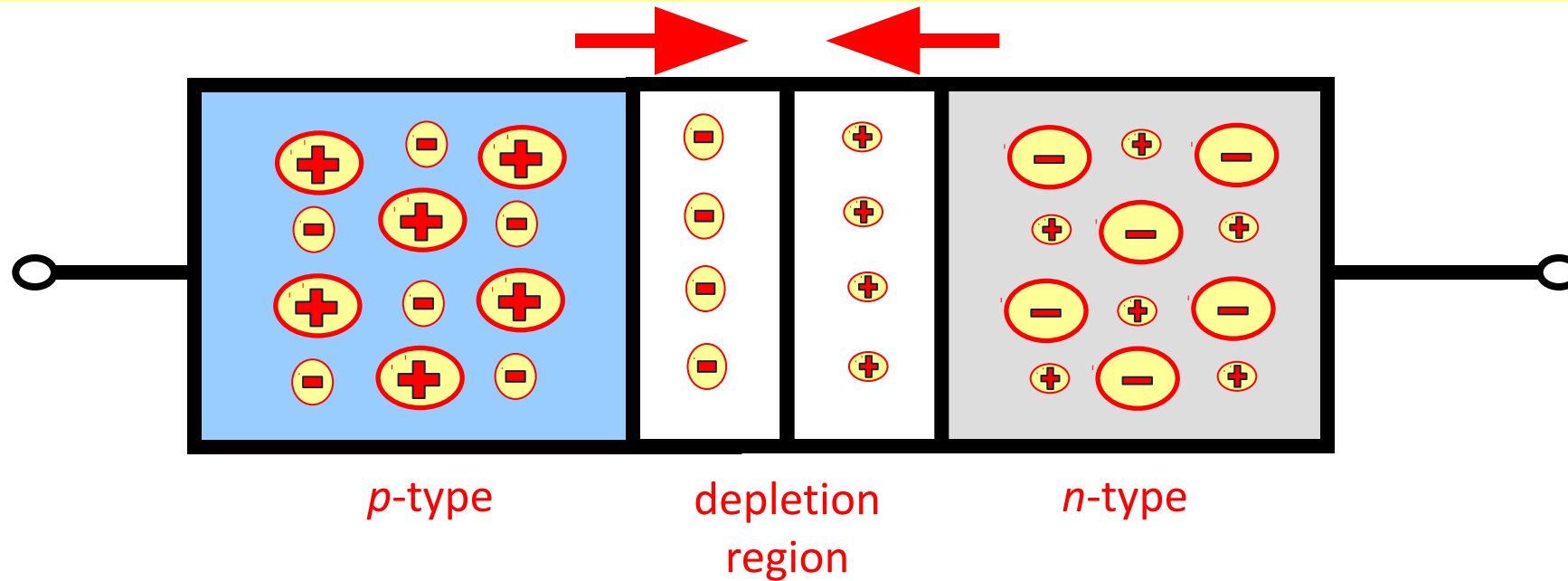


Figure: The *pn* junction with no applied voltage (open-circuited terminals).

»»» pn Junction

Step #6: Equilibrium is reached, and diffusion ceases, once the magnitudes of diffusion and drift currents equal one another – resulting in **no net flow**.

Once equilibrium is achieved, no net current flow exists ($I_{net} = I_D - I_S$) within the *pn*-junction while under open-circuit condition.



»»» The Drift Current I_s and Equilibrium

- In addition to majority-carrier diffusion current (I_D), a component of **current due to minority carrier drift exists (I_s)**.
- Specifically, some of the **thermally generated holes** in the p -type and n -type materials move toward and reach the edge of the depletion region.
- There, they experience the electric field (V_o) in the depletion region and are **swept across it**.
 - Unlike diffusion current, the polarity of V_o **reinforces** this drift current.

»»» Operation with Open-Circuit Terminals

- **Q:** What has been learned about the pn -junction?
 - **A: composition**
 - The pn junction is composed of two silicon-based semiconductors, one doped to be p -type and the other n -type.
 - **A: majority carriers**
 - Are generated by doping.
 - Holes are present on p -side, free electrons are present on n -side.

»»» Operation with Open-Circuit Terminals

- **Q:** What has been learned about the *pn*-junction?
 - **A:** bound charges
 - Charge of majority carriers are **neutralized electrically by bound charges.**
 - **A:** diffusion current I_D
 - Those majority carriers close to the junction will **diffuse across, resulting in their elimination.**

»»» Operation with Open-Circuit Terminals

- **Q:** What has been learned about the pn-junction?
 - **A:** depletion region
 - As these carriers disappear, they **release bound charges** and effect a voltage differential V_0 .
 - **A:** depletion-layer voltage
 - As **diffusion continues, the depletion layer voltage (V_0) grows**, making diffusion more difficult and eventually bringing it to halt.

»»» Operation with Open-Circuit Terminals

- **Q:** What has been learned about the pn-junction?
 - **A:** minority carriers
 - Are **generated thermally**.
 - Free electrons are present on p -side, holes are present on n -side.
 - **A:** drift current I_s
 - The depletion-layer voltage (V_0) facilitates the flow of minority carriers to opposite side.
 - **A:** open circuit equilibrium $I_D = I_s$

Qualitative Description of Junction Operation

- Figure to right shows pn -junction under three conditions:
 - (a) **open-circuit** – where a barrier voltage V_0 exists.
 - (b) **reverse bias** – where a dc voltage V_R is applied.
 - (c) **forward bias** – where a dc voltage V_F is applied.

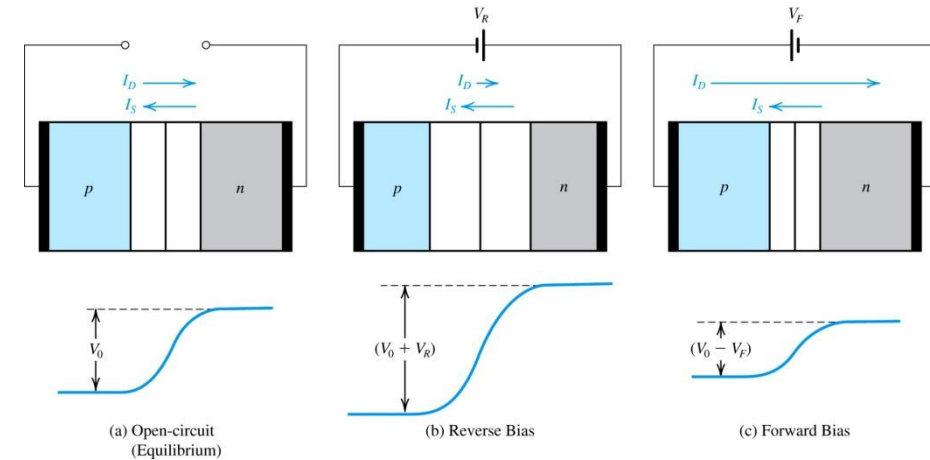


Figure 3.11: The pn junction in: (a) equilibrium; (b) reverse bias; (c) forward bias.

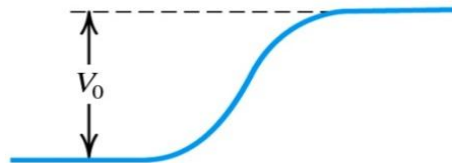
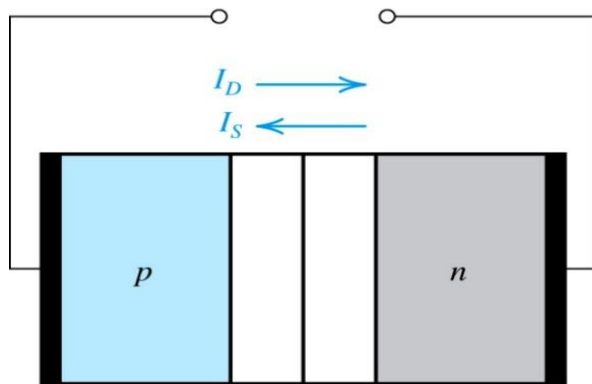


Qualitative Description of Junction Operation

1) no voltage applied

2) voltage differential across depletion zone is V_0

3) $I_D = I_S$

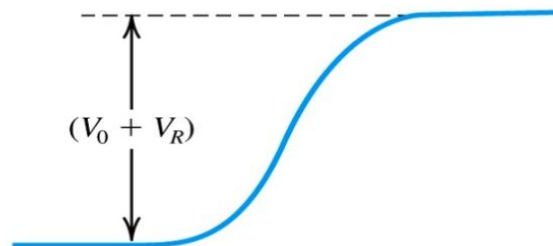
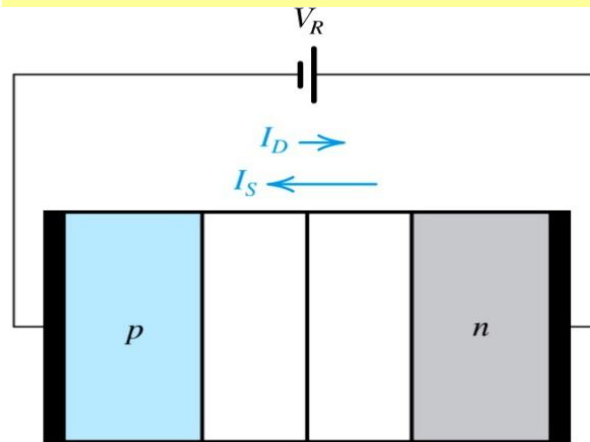


(a) Open-circuit
(Equilibrium)

1) negative voltage applied

2) voltage differential across depletion zone is $V_0 + V_R$

3) $I_D < I_S$

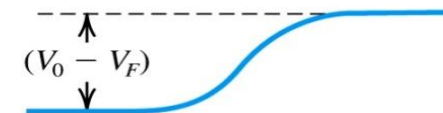
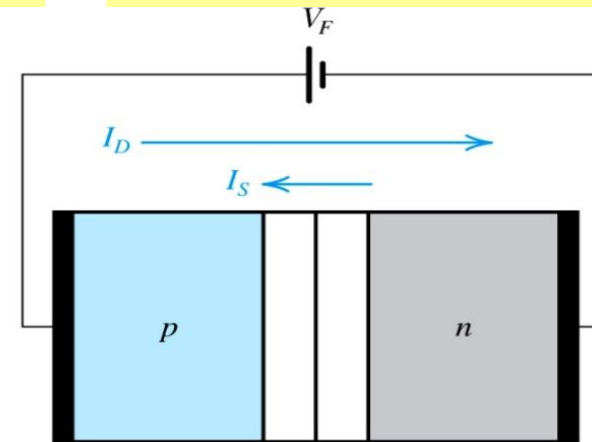


(b) Reverse Bias

1) positive voltage applied

2) voltage differential across depletion zone is $V_0 - V_F$

3) $I_D > I_S$



(c) Forward Bias



Qualitative Description of Junction Operation

■ reverse bias case

- the externally applied voltage V_R **adds to (aka. reinforces) the barrier** voltage V_0
 - ...increase effective barrier
- this **reduces rate of diffusion**, reducing I_D
 - if $V_R > 1V$, I_D will fall to **0A**
- the drift current I_S **is unaffected**, but dependent on temperature
- result is that *pn* junction will **conduct small drift current I_S**

minimal current flows in reverse-bias case

■ forward bias case

- the externally applied voltage V_F **subtracts from the barrier** voltage V_0
 - ...decrease effective barrier
- this **increases rate of diffusion**, increasing I_D
- the drift current I_S **is unaffected**, but dependent on temperature
- result is that *pn* junction will **conduct significant current $I_D - I_S$**

significant current flows in forward-bias case

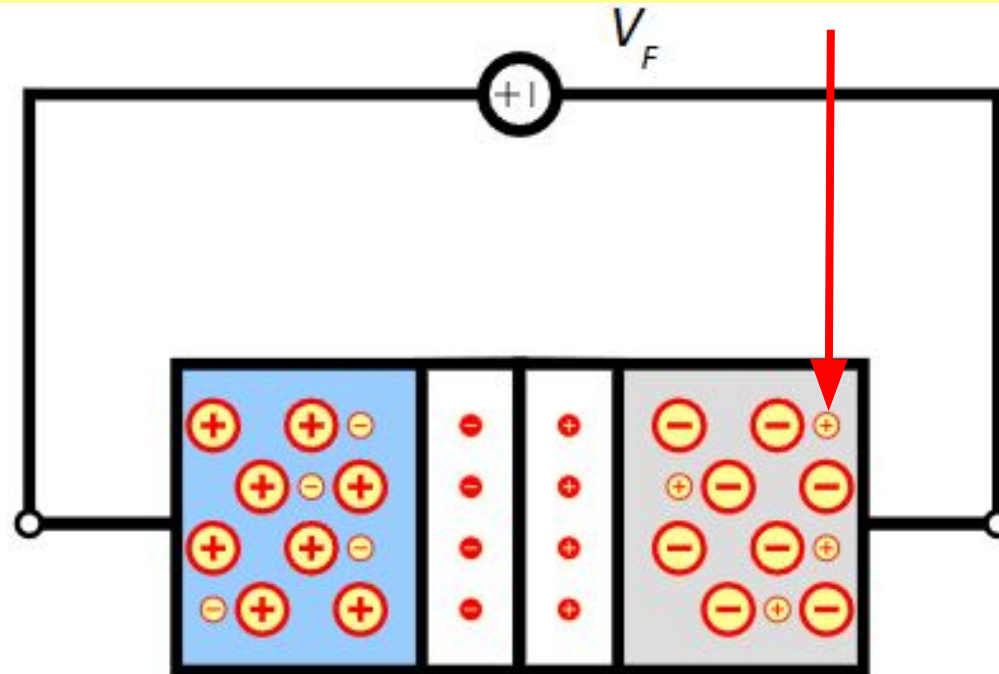
»» The Current-Voltage Relationship of the Junction

- **Q:** What happens, exactly, when a **forward-bias voltage** (V_F) is applied to the *pn*-junction?
 - **step #1:** Initially, a small forward-bias voltage (V_F) is applied. It, because of its polarity, **pushes majority carriers (holes in *p*-region and electrons in *n*-region) toward the junction** and reduces width of the depletion zone.
 - Note, however, that this **force is opposed by the built-in voltage** built in voltage V_0 .

»»» The Current-Voltage Relationship of the Junction

step #1: Initially, a small forward-bias voltage (V_F) is applied. It, because of its polarity, **pushes majority (holes in p -region and electrons in n -region) toward the junction** and reduces width of the depletion zone.

Note that, in this figure, the smaller circles represent minority carriers and **not bound charges** – which are not considered here.



»»» The Current-Voltage Relationship of the Junction

step #2: As the magnitude of V_F increases, the depletion zone becomes thin enough such that the barrier voltage ($V_0 - V_F$) cannot stop diffusion current – as described in previous slides.

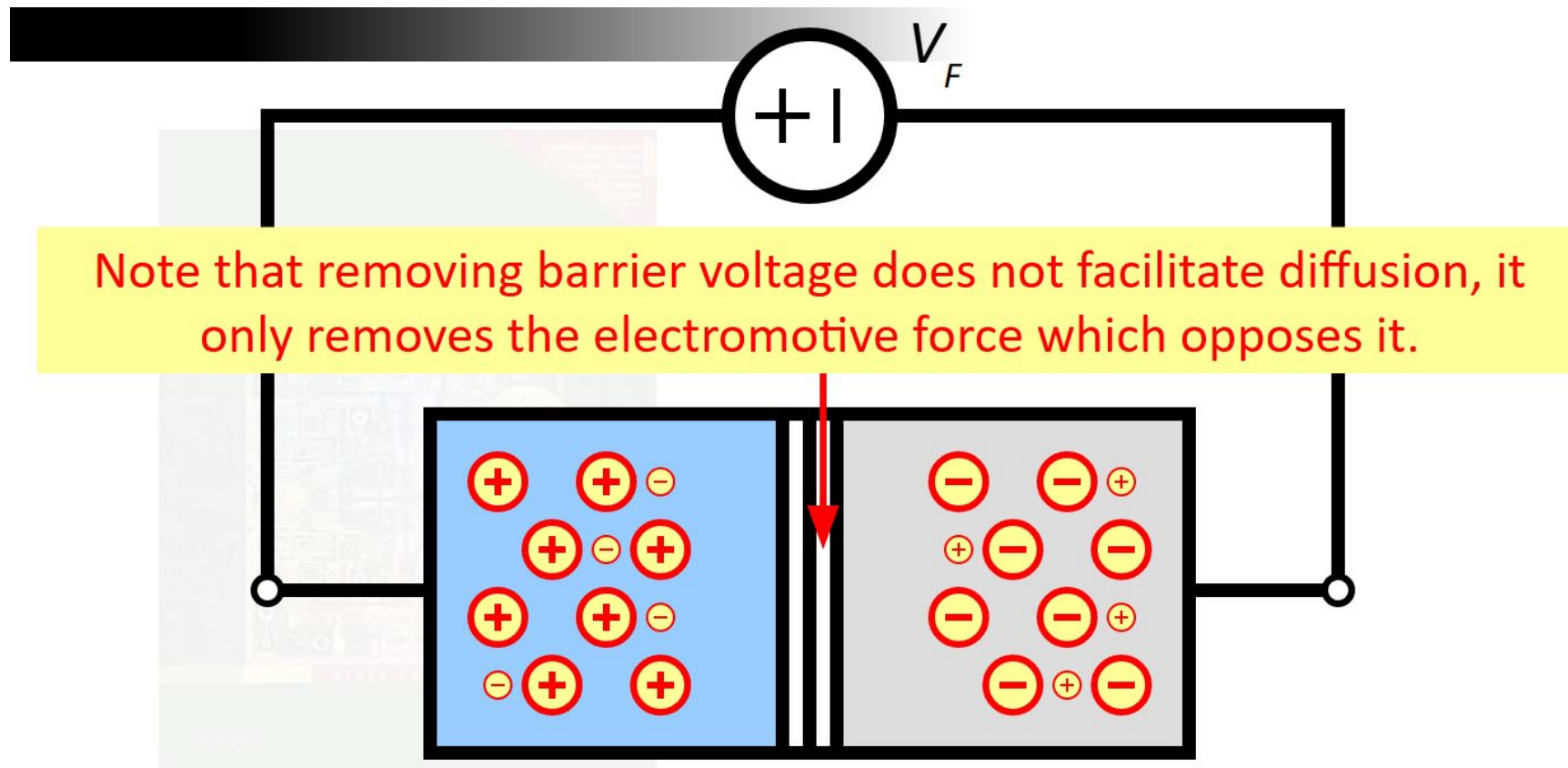


Figure: The *pn* junction with applied voltage.

»»» The Current-Voltage Relationship of the Junction

step #3: Majority carriers (free electrons in n -region and holes in p -region) **cross the junction and become minority charge carriers** in the near-neutral region.

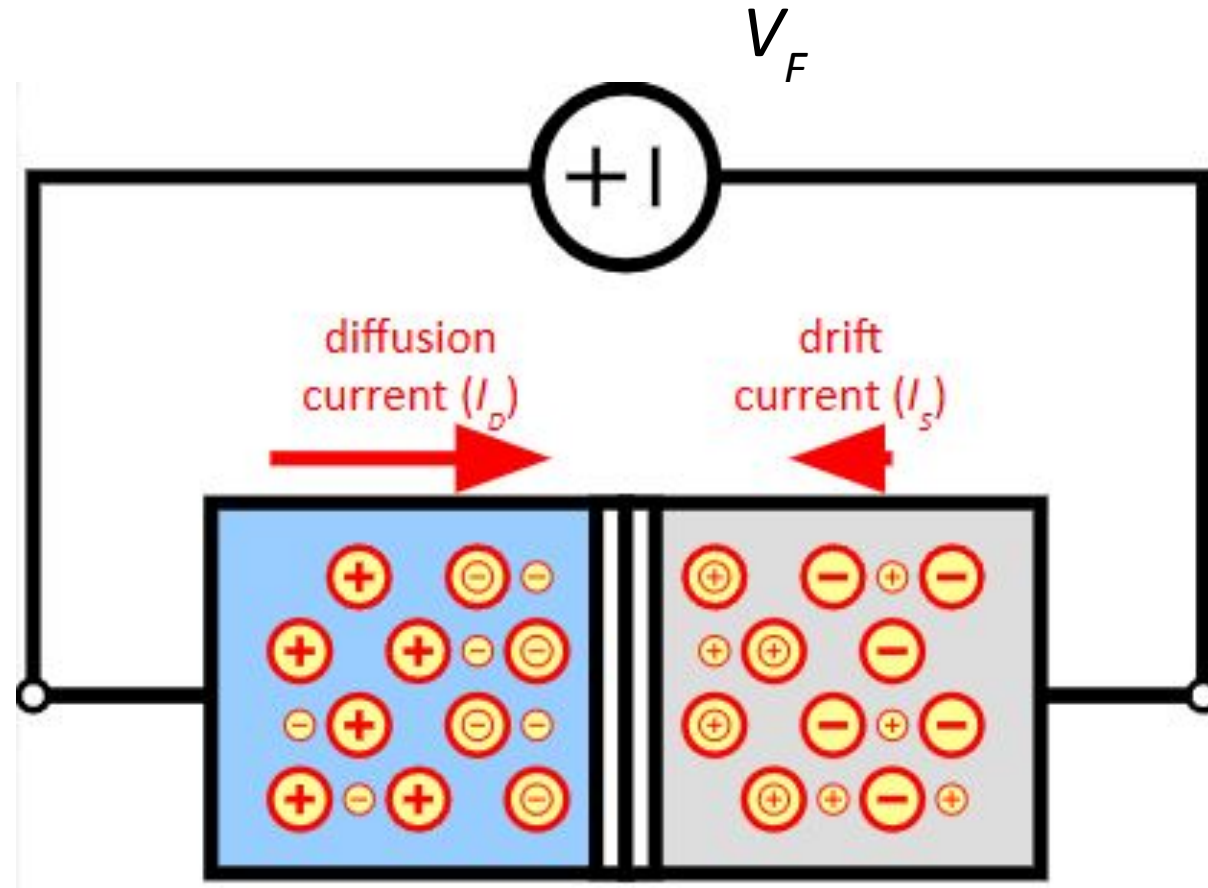


Figure: The pn junction with applied voltage.

»» The Current-Voltage Relationship of the Junction

$$(eq3.40) \quad I = I_s (e^{V/V_T} - 1)$$

- **saturation current (I_s)** – is the **maximum reverse current** which will flow through *pn*-junction.
 - It is proportional to **cross-section of junction (A)**.
 - Typical value is 10^{-18} **A**.

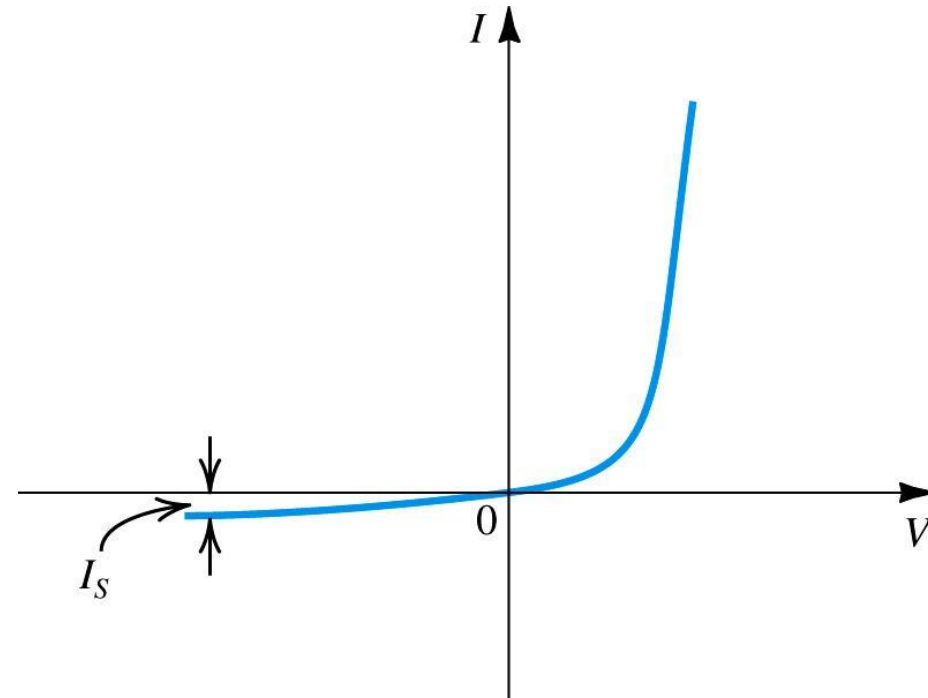


Figure 3.13: The *pn* junction I – V characteristic.